

Preliminary Proposal ,Week 4 Refined

Project Title: AI-Assisted Triage Support Tool for High-Risk Patient Identification in the Emergency Department

Student: Mithra Ramoutar | Supervisor: Dr. L. De Freitas

Statement of the Problem

Mercer General Hospital's ED faces risk of patient under-triage during peak periods, where nurse workload and time pressure can cause high-risk cases to be missed. A 12-week pilot will test an explainable AI support tool that uses structured triage data and free-text presenting complaints to flag high-risk cases for nurse review, displaying a visible reason for each alert. Nurses retain full accountability for all final escalation decisions.

Previous Work

A growing body of literature has examined artificial intelligence in emergency department triage and clinical AI safety. The fifteen studies below form the evidence base for this proposal.

Tyler et al. (2024)

A scoping review found that AI and machine-learning tools show promise for acuity prediction, resource allocation, and earlier recognition of urgent cases, while noting that most evidence remains retrospective and limits conclusions about real-world effectiveness and safety.

Da'Costa et al. (2025)

This review highlights potential improvements in triage speed and consistency, but identifies bias, data quality, explainability, privacy, and clinician trust as major implementation concerns ,a gap directly relevant to this proposal.

Araouchi & Adda (2024)

TriageIntelli used multimodal triage data and achieved strong KTAS classification performance (AUROC 0.93), but its single-dataset design limits generalisability across different clinical environments and demographics.

Chen et al. (2024)

Machine-learning and NLP models using structured triage data and free-text notes improved disposition prediction beyond physician baselines, though longer-term outcomes and resource use were not fully addressed.

Ottenhoff et al. (2023)

Machine learning may support more consistent triage and identify patients at risk of adverse outcomes, but live clinician trust and workflow impact remain unresolved in real ED settings.

Townsend et al. (2023)

Practitioners saw value in AI support but raised concerns about accountability, transparency, bias, and over-reliance, demonstrating that implementation challenges extend well beyond technical performance.

Hinson et al. (2024)

A multisite AI-informed decision-support implementation improved triage equity across demographic groups, providing a real-world precedent for fairness-focused AI deployment in emergency triage settings.

De Freitas et al. (2020)

This Caribbean ED patient-flow study provides the regional context for workflow mapping, including CTAS usage, resource constraints, and high attendance volumes unique to developing health systems in Trinidad and Tobago.

Cheng et al. (2024)

Implementation facilitators include staff education, efficient workflow integration, and external validation; barriers include clinician resistance and poor change management ,findings that directly inform this pilot's deployment strategy.

Zayas-Caban et al. (2023)

This paper identifies data quality, EHR integration, clinician trust, and regulatory readiness as key barriers to AI adoption in emergency care, recommending iterative co-design with clinical staff.

Wong et al. (2021)

An external validation of the Epic Sepsis Model across 38,455 hospitalisations found a positive predictive value of just 12.8% and a sensitivity of 33%, despite widespread deployment. This case study is the primary reference for distribution shift and automation bias risks in this proposal's risk register.

Obermeyer et al. (2019)

Obermeyer et al. demonstrated that a widely used commercial algorithm for managing population health systematically discriminated against Black patients by using health cost as a proxy for health need. This landmark study establishes the theoretical foundation for the equity risks identified in this proposal.

Kumar et al. (2023)

This review of AI ethics and trust challenges in healthcare identifies transparency, accountability, and explainability as the three most consistently cited ethical requirements for clinical AI deployment ,informing the alert design decisions in this pilot.

Choudhury & Asan (2021)

This systematic review examines how AI safety is defined and measured in healthcare, finding that most studies report technical performance metrics while neglecting human factors, workflow integration, and unintended consequences in live clinical environments.

McCradden et al. (2020)

McCradden et al. propose ethical principles for a regulatory approach to bias in healthcare machine learning, arguing that bias assessment must be embedded throughout the model lifecycle ,from training data audit through post-deployment monitoring ,rather than treated as a one-time pre-launch check.

Gaps in the Literature

- Many studies report retrospective model performance but fewer evaluate whether AI improves live nurse triage decisions in real ED workflows. This pilot addresses this by embedding the tool directly into the triage process.
- Accuracy metrics are common, while alert burden, explainability, demographic fairness, and clinician actionability are less consistently measured. This pilot treats all four as primary KPIs.
- No identified study evaluates AI triage performance in a Caribbean ED context, where local workflow constraints, documentation practices, and patient demographics may significantly affect model performance and clinician adoption.
- Risk registers and prospective safety frameworks are rarely published alongside AI triage proposals. This proposal addresses that gap by including a structured risk register with mitigations and measurable signals of success.

Proposed Solution

This pilot proposes an explainable AI support tool for emergency department triage at Mercer General Hospital. It will process structured triage inputs, vital signs, chief complaints, and acuity scores, alongside free-text presenting complaint notes to identify patients at elevated risk of under-triage.

The model will combine gradient boosting and natural language processing to flag high-risk patients for nurse review. Each alert will display a human-readable reason, such as elevated heart rate combined with complaint keywords associated with sepsis, so nurses can evaluate or override the recommendation with full clinical accountability.

The tool integrates at the triage nurse assessment stage, following vitals capture and preceding CTAS assignment. Three workflow constraints shape its design: EHR data lag, nurse cognitive load and alert fatigue, and free-text documentation variability. These informed the one-glance alert design, high-confidence-only flagging, and asynchronous EHR data tolerance.

The 12-week pilot will measure:

- Sensitivity and specificity of high-risk flagging compared to retrospective clinical outcomes
- Alert burden (flags per shift) to ensure the tool does not overwhelm triage nurses
- Fairness across demographic groups to detect and mitigate potential bias in predictions
- Effect on escalation decision time against baseline triage workflow data

ED Triage Workflow and AI Integration Points

The tool sits at AI point 3 in the ED triage workflow: triage nurse assessment, following vitals capture and preceding CTAS level assignment. The full annotated workflow diagram is committed to `carisurg-portfolio/docs/week3_triage_flow.md`. Five AI integration points are mapped: door wait prediction (point 1), vitals anomaly detection (point 2), NLP risk scoring at nurse assessment (point 3, core), CTAS consistency check at physician assessment (point 4), and post-discharge deterioration flagging at disposition (point 5).

Constraints and Stakeholders

Constraint 1: EHR data lag

Vitals are recorded on paper before EHR entry, introducing a 5-30 minute lag. The tool must interface with bedside monitoring or tolerate delayed data with explicit uncertainty flagging.

Constraint 2: Nurse cognitive load and alert fatigue

Alerts must be rare, high-confidence, and readable at a glance. Alert burden per shift is a primary KPI. One-glance design is mandatory: alert, reason, and action in under three seconds.

Constraint 3: Free-text documentation variability

The NLP component must be calibrated on locally sourced Mercer ED triage notes before deployment. Models trained on clean clinical corpora will underperform on abbreviated, colloquial, or locally inflected text.

Clinical Stakeholders

Stakeholder	Role	Top concern
Triage Nurse (Sr. Patrice Alleyne)	First clinical contact; assigns CTAS level	Will this slow me down or create extra steps during a busy shift?

ED Consultant Physician	Final clinical authority; disposition decisions	Is the AI accountable, auditable, and covered by clinical governance?
Clinical IT Lead (Martina Griffith)	Oversees system integration and audit	Does this meet data governance, privacy, and security standards?
Patient / Patient Advocate	Recipient of triage decision	Will I be treated fairly regardless of background, language, or insurance status?
Ward Porter / Healthcare Assistant	Transfers patients between areas	Will alert-driven escalations increase my workload unpredictably?

Risk Analysis

This section presents the risk register and risk memo produced in response to the Saint Cedric Ministry of Health's consultation on a nationwide rollout of AI-assisted triage. Risks are grouped across four dimensions: AI-technical, operational, ethical, and equity.

Risk Memo

Verdict

The proposed AI triage support system carries manageable risk if three conditions are met: the model is validated on locally sourced Caribbean ED data before deployment, alert burden is actively monitored and kept below five flags per shift, and the alert UI makes clinical override cognitively easy rather than requiring justification.

Top 3 Risks

- Distribution shift ,the model will almost certainly be trained on data from non-Caribbean populations. Deployed at Mercer without local validation, it may systematically underperform on the patient demographics it is intended to serve.
- Alert fatigue ,a tool that generates too many low-confidence alerts will be ignored within weeks of go-live. This is the failure mode most likely to produce patient harm through a false sense of AI-assisted safety.
- Automation bias ,nurses presented with an AI recommendation may defer to it even when their clinical judgement differs. This risk is highest in high-volume, high-fatigue conditions ,exactly when the tool is most likely to be used.

Mitigations

- Distribution shift: fine-tune the model on a minimum of 500 locally sourced Mercer ED encounters before go-live; repeat validation quarterly.
- Alert fatigue: set alert threshold at >0.85 model confidence; cap alerts at five per shift; review threshold weekly during the first month of the pilot.
- Automation bias: design the UI to present the AI score as one input among several; require a one-tap override with no justification required; conduct pre-launch simulation training with Sr. Patrice Alleyne's team.

Signals of Success

- Distribution shift mitigation: sensitivity and specificity on local validation data are within one standard deviation of published benchmarks.
- Alert fatigue mitigation: alert override rate remains below 30% at four weeks; nurses rate the tool as 'helpful' or 'very helpful' on weekly usability survey.
- Automation bias mitigation: nurses correctly override the tool in at least 80% of injected test scenarios where the AI recommendation is deliberately incorrect.

Residual Risk

One risk cannot be fully resolved at this stage: the absence of a Caribbean ED training dataset means the model's baseline fairness across the specific demographic profile of Mercer's patient population cannot be established before pilot launch. This is an honest limitation. It will be addressed through prospective fairness monitoring during the pilot, with a pre-defined threshold that triggers tool suspension if demographic performance gaps exceed 5%.

Full Risk Register

Eight named risks across four categories. L = Low, M = Medium, H = High.

Risk Name	Category	L'ho o d	Impa c t	Mitigation	Signal of Success
Demographic bias in training data	AI-Technical	H	H	Audit training data distribution; apply re-weighting or stratified sampling.	Sensitivity/specificity equivalent across demographic groups in pilot evaluation.
Distribution shift , non-Caribbean training data	AI-Technical	H	H	Fine-tune on minimum 500 local Mercer ED encounters; retrain quarterly.	Local validation performance within one standard deviation of benchmarks.
NLP degradation on abbreviated free-text	AI-Technical	M	H	Calibrate NLP on representative sample of real Mercer triage notes before go-live.	NLP accuracy on local notes within 5% of accuracy on clean clinical corpus.
Alert fatigue , clinician disengagement	Operational	H	H	Threshold >0.85 confidence; cap at 5 alerts/shift; weekly calibration review.	Override rate below 30% at 4 weeks; nurses rate tool as helpful on usability survey.
EHR data lag ,stale inputs at point of alert	Operational	H	M	Operate on triage card data, not EHR confirmation; display data-age indicator.	Fewer than 10% of alerts triggered on inputs more than 15 minutes old.
Automation bias , deference to AI over judgement	Ethical	M	H	Pre-launch training on AI limitations; UI shows alert as advisory; log all overrides.	Nurses override correctly in 80%+ of injected incorrect-recommendation scenarios.
Lack of informed patient consent	Ethical	M	M	Visible notice in waiting area; ethics board clearance before pilot launch.	Ethics board approval obtained; zero patient complaints related to AI use in pilot.
Underperformance on paediatric and elderly patients	Equity	M	H	Stratify evaluation by age; suspend tool for affected cohorts if gap exceeds 5%.	Sensitivity gap between adult and paediatric/elderly cohorts below 5%.

Real-World AI Harm Case Study

Incident: Epic Sepsis Model ,University of Michigan Health System (2021)

What happened

The Epic Sepsis Model (ESM) is a machine-learning algorithm embedded within the Epic electronic health record system and deployed at hundreds of US hospitals to flag patients at risk of developing sepsis. A 2021 independent evaluation published in JAMA Internal Medicine by Wong et al. audited the ESM across 38,455 patient encounters at the University of Michigan Health System. The study found a positive predictive value of just 12.8% ,fewer than 1 in 8 flagged patients actually developed sepsis ,and a sensitivity of 33.4%, meaning the model missed two-thirds of patients who did develop the condition. The tool was in active clinical use, generating alerts nurses and physicians were expected to act on.

Root cause analysis

- Training-deployment distribution shift. The model was trained on retrospective data from a single health system (University of Wisconsin) and deployed without independent external validation across institutions with different patient populations and documentation practices.
- Opaque validation reporting. Epic did not publish the model's source code, training data, or full performance statistics, leaving hospitals with no basis to evaluate suitability for their patient population before go-live.
- Automation bias in clinical response. Despite poor specificity, clinical teams reported pressure to act on ESM alerts because they were embedded in EHR workflows and associated with institutional sepsis protocols, making override feel like inaction.
- No signal-of-success monitoring. Hospitals had no pre-defined criteria for when the model was underperforming in their specific context and no mechanism to trigger a review or suspension of the tool.

Relevance to the Mercer pilot

The Epic Sepsis Model case is the clearest published precedent for the risks in this proposal. Three direct design responses follow. First, the Mercer tool must be validated on locally sourced ED data before deployment ,external validation statistics cannot be assumed to transfer. Second, performance must be monitored continuously during the pilot, with pre-defined thresholds triggering automatic review if sensitivity or specificity fall below acceptable levels. Third, the alert UI must make override cognitively easy. The ESM case confirms that distribution shift and automation bias are the two highest-priority risks in this register, both rated High likelihood and High impact.

Source

Wong, A., Otles, E., Donnelly, J. P., Krumm, A., McCullough, J., DeTroyer-Cooley, O., Pestrue, J., Phillips, M., Konye, J., Penzoza, C., Ghous, M., & Singh, K. (2021). External validation of a widely implemented proprietary sepsis prediction model in hospitalized patients. JAMA Internal Medicine, 181(8), 1065-1070. <https://doi.org/10.1001/jamainternmed.2021.2626>

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